

Advanced Statistical Methods

Non-Parametric Statistics

Complete Lecture Notes

Weeks 01 04

Hypothesis Testing · Estimation · Rank Methods

CDF Inference · Multiple Samples

Sign Test · Wilcoxon Signed Rank · Wilcoxon Rank Sum
Mann-Whitney · Kruskal-Wallis · Kolmogorov-Smirnov

Distribution-Free Methods for Statistical Inference

Non-Parametric Statistics: Complete Lecture Notes

Advanced Statistical Methods (ASM-I)

Weeks 0104

These notes compile lectures on non-parametric and distribution-free methods for hypothesis testing and estimation.

Compiled from lecture slides, handouts, and class notes.

All mathematical content is presented for educational purposes.

Contents

1	Introduction to Non-Parametric Methods	5
1.1	Recap: Parametric Methods	5
1.1.1	One-Sample Mean	5
1.1.2	Two-Sample Mean	5
1.1.3	Variance Test (One Sample)	6
1.1.4	Two-Sample Variance	6
1.1.5	ANOVA (One-Way)	6
1.2	Motivation for Non-Parametric Tests	6
1.2.1	Why Not Always Use Non-Parametric Methods?	7
1.3	The Sign Test (One-Sample Median Test)	7
1.3.1	Concept	7
1.3.2	Example Scenario: Arsenic Concentration	8
1.3.3	Procedure	8
1.3.4	Rejection Region	8
1.3.5	Sign Test Analysis	8
1.3.6	Large-Sample Approximation for the Sign Test	9
1.3.7	Summary of the Sign Test	9
1.4	Wilcoxon Signed Rank Test	9
1.4.1	Procedure	10
1.4.2	Worked Example: Heart Rate	10
1.4.3	Distribution of T	10
1.4.4	Exact Null Distribution	11
1.4.5	Comparison of Tests	11
1.5	Large-Sample Approximation for the Signed Rank Test	12
1.5.1	Derivation of Mean and Variance	12
1.6	Handling Ties	13
1.6.1	Case 1: Zeros	13
1.6.2	Case 2: Equal Absolute Differences	13
1.7	R Code: Week 1	14
2	One-Sample Estimation and Two-Sample Tests	15
2.1	One-Sample Estimation of Centre	15
2.1.1	Derivation of Interval Estimates	15
2.1.2	Example: Academic Salary Data	16
2.2	Two-Sample Test for Centre	16
2.2.1	Setup	16
2.3	Wilcoxon Rank Sum Test	17
2.3.1	Procedure	17
2.3.2	Numerical Example: Reaction Times	18
2.3.3	Null Distribution Construction	18
2.4	Permutation Tests	19

2.4.1	Application to the t -Statistic	19
2.5	R Code: Week 2	19
3	Mann-Whitney U-Statistic, General Score Functions, and Scale Tests	21
3.1	Wilcoxon Rank Sum and Mann-Whitney Equivalence	21
3.1.1	Properties of the U-Statistic	21
3.1.2	Asymptotic Normality	22
3.2	General Linear Rank Statistics	22
3.2.1	General Form	22
3.2.2	Assumptions on the Score Function φ	22
3.3	Measure of Scale	23
3.4	Test for Comparing Scales of Two Populations	23
3.4.1	Setup	23
3.4.2	Fligner-Killeen (FK) Test	24
3.4.3	Numerical Result	24
4	Multiple Samples, CDF Inference, and Goodness of Fit	27
4.1	Multiple Samples	27
4.1.1	Parametric Case	27
4.1.2	Non-Parametric Case	27
4.2	Kruskal-Wallis Test	28
4.3	Inference for the CDF	29
4.3.1	Empirical CDF	29
4.3.2	Order Statistics and the Empirical CDF	30
4.3.3	Testing / Estimation of Median	30
4.4	Confidence Intervals for CDF and Quantile Functions	31
4.5	Testing for Median (L-6)	31
4.5.1	Is the Population Median 0.2?	31
4.5.2	Is the Third Quartile 0.3?	31
4.6	Estimating the CDF Further Details	32
4.7	Goodness of Fit: χ^2 and KS Tests	33
4.7.1	Case 1: F_0 Discrete χ^2 Test	33
4.7.2	Kolmogorov-Smirnov Statistic	33
4.7.3	Distribution-Free Property of D	33
5	Density Estimation	35
5.1	Density Estimation	35

Chapter 1

Introduction to Non-Parametric Methods

1.1 Recap: Parametric Methods

Before introducing non-parametric methods, we briefly review the standard parametric tests that motivate the need for distribution-free alternatives.

1.1.1 One-Sample Mean

- **Assumed Model:** $\mathbb{E}[X_i] = \mu$
- **Hypothesis Tests:**

$$H_0 : \mu = \mu_0 \quad H_a : \mu \neq \mu_0$$

- **Test Statistic:**

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

- **Assumptions:**

- $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$
- If n is large, F can be any distribution (CLT applies).
- If n is small, we assume $F \sim \mathcal{N}$ (Normal Distribution).

- **Open question:** What if n is small and F is not Normal?

1.1.2 Two-Sample Mean

- **Samples:** $X_1, \dots, X_n \sim F$ and $Y_1, \dots, Y_m \sim G$
- **Assumption:** $\mathbb{E}[X] = \mu_1, \mathbb{E}[Y] = \mu_2$
- **Hypothesis:**

$$H_0 : \mu_1 = \mu_2 \quad H_a : \mu_1 \neq \mu_2$$

- **Conditions:**

- If F and G are both Normal $\Rightarrow$ 2-sample t -test.
- If both m and n are large $\Rightarrow$ Normal test.

1.1.3 Variance Test (One Sample)

• **Sample:** $X_1, \dots, X_n \sim F$

• **Variance:** $\text{Var}(X_i) = \sigma^2$

• **Statistic:**

$$\frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$$

• S^2 is the unbiased sample variance.

• **Hypothesis:** $H_a : \sigma^2 \neq \sigma_0^2$

• **Assumption:** F is Normal.

1.1.4 Two-Sample Variance

• **Statistic:** S_1^2/S_2^2

• **Hypothesis:** $H_0 : \sigma_1^2 = \sigma_2^2$

• **Test:** F -test

• **Assumption:** $F = \mathcal{N} = G$ (both distributions are Normal).

1.1.5 ANOVA (One-Way)

• Tests equality of means of several samples.

• Valid under Normal assumptions.

1.2 Motivation for Non-Parametric Tests

Remark

Central question: What if the sample size n is small and the parent distribution is unknown?

• If the parent distribution is *known* but not Normal, we may be able to construct the distribution of the test statistic using probability theory. For example:

$$- F \sim \text{Exp}(\lambda) \Rightarrow \sum X_i \sim \text{Gamma}(n, \lambda)$$

$$- F \sim \text{Pois}(\lambda) \Rightarrow \sum X_i \sim \text{Pois}(n\lambda)$$

Then we can construct the test statistic under H_0 using knowledge of probability theory.

• However, *often we do not know the distribution.*

Definition: Distribution-Free Tests

The tests we shall construct are **distribution-free**. We do not assume any parametric model a model with known form but a finite number of unknown parameters.

1.2.1 Why Not Always Use Non-Parametric Methods?

The flexibility of non-parametric methods is attractive but comes with a cost:

- When a parametric model is true/correct, using that parametric test is “better”.
- **Power:** For hypothesis testing, both parametric and non-parametric tests will have the same significance level α . But the parametric test (under the correct parent model) will have more power.
- **Confidence Intervals:** Both procedures will give the same average probability (coverage), but the parametric interval will be shorter/narrower (more precise).

1.3 The Sign Test (One-Sample Median Test)**Remark**

Use non-parametric techniques if the distribution of the data is unknown. The Sign Test is the non-parametric alternative to the one-sample t -test.

1.3.1 Concept

- Generally for F , the mean may not exist (e.g., Cauchy distribution), but the **Median always exists**.
- Median equals Mean if $F \approx \mathcal{N}$ (Normal).

Definition: Median

The **median** M of a distribution F is defined as:

$$M = \min\{a : F(a) \geq \frac{1}{2}\}.$$

1.3.2 Example Scenario: Arsenic Concentration

Example: Arsenic in Drinking Water

Source: StatLib (CMU) Arsenic in drinking water.

Data: $X_1, \dots, X_{21}$ (Arsenic concentration in toenail clippings of 21 individuals).

Hypothesis:

$$H_0 : \theta = 0.2 \quad \text{vs} \quad H_a : \theta > 0.2 \quad (\text{Right-tailed})$$

where $\text{med}(X_i) = \theta$.

1.3.3 Procedure

Procedure: Sign Test Procedure

1. Under H_0 , observe that $P(X_i > 0.2) = \frac{1}{2}$.
2. Sample $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$ (independent).
3. Define the indicator variable Y_i :

$$Y_i = \begin{cases} 1 & \text{if } X_i > \theta_0 \\ 0 & \text{if } X_i \leq \theta_0 \end{cases}$$

4. Test Statistic:

$$Y = \sum_{i=1}^n Y_i \sim \text{Bin}(n, \frac{1}{2})$$

(Since X_i are independent, Y_i are independent Bernoulli trials.)

1.3.4 Rejection Region

- Reject for *large* values of Y .
- $P(Y > C) = \alpha$ (Right-tailed).
- Depending on the type of H_a , a left-tailed or two-tailed rejection region may be needed.

1.3.5 Sign Test Analysis

Remark

A **randomness check** is necessary before applying the sign test.

Example: Sign Test Calculation

$H_a : \theta > 0.2$, observed $y = 10$ with $n = 21$.

P-value calculation (in R):

```
1 2.pbinom(10, 21, 0.5)    # gives 1
```

If p-value $> \alpha$, do *not* reject H_0 .

1.3.6 Large-Sample Approximation for the Sign Test

- If n is large, use the Normal approximation to the Binomial (converges to normal quickly; $n \geq 10$ is usually sufficient). Note: convergence to a general F may require a much larger n .
- Test Statistic: $Y \sim \text{Bin}(n, 0.5)$.
- With $n = 21$, $y = 10$:

$$Z = \frac{Y - np}{\sqrt{npq}} = \frac{Y - 0.5n}{0.5\sqrt{n}}$$

- In R:

```
1 2 * pnorm(Z)    # gives 0.8
```

- Do not reject H_0 at any $\alpha < 0.8$.

1.3.7 Summary of the Sign Test**Theorem: Sign Test Summary**

- A one-sample median test looking at the *sign* of $X_i - \theta_0$.
- Widely applicable. The only assumption is that F is **continuous**.
- **Drawback:** Low power under many alternatives. It ignores the *magnitude* of the difference.
- Example: A density could be symmetric but shaped such that signs are balanced even if the bulk of data is far from θ_0 .

1.4 Wilcoxon Signed Rank Test**Definition: Wilcoxon Signed Rank Test**

- Assumes F is **continuous and symmetric**.
- Ranks are distribution-free.
- **Idea:** Give more weight to X_i that are much larger than θ_0 (large deviations receive larger weight).

1.4.1 Procedure

Procedure: Wilcoxon Signed Rank Procedure

1. Calculate absolute differences: $|X_i - \theta_0|$.
2. Assign ranks R_i to these absolute differences (rank 1 = smallest).
3. Maximum sum of ranks: $\sum R_i = \frac{n(n+1)}{2}$.
4. Test Statistic:

$$T = \sum R_i Y_i$$

where Y_i is the indicator of positive sign (as defined for the Sign Test):

$$Y_i = \begin{cases} 1 & \text{if } X_i > \theta_0 \\ 0 & \text{if } X_i \leq \theta_0 \end{cases}$$

1.4.2 Worked Example: Heart Rate

Example: Wilcoxon Signed Rank Heart Rate Data

$$H_0 : \text{Median} = 70 \quad \text{vs} \quad H_a : \text{Median} > 70, \quad \theta_0 = 70.$$

Data (X_i): 73, 82, 87, 68, 106, 60, 97.

X_i	$X_i - \theta_0$	Sign (Y_i)	$ X_i - \theta_0 $	Rank (R_i)	$R_i Y_i$
73	+3	1	3	2	2
82	+12	1	12	4	4
87	+17	1	17	5	5
68	-2	0	2	1	0
106	+36	1	36	7	7
60	-10	0	10	3	0
97	+27	1	27	6	6

Test Statistic:

$$T = \sum R_i Y_i = 2 + 4 + 5 + 7 + 6 = 24.$$

1.4.3 Distribution of T

- Let $Y = \sum_{i=1}^n Y_i$ (number of positive signs).
- $T = \sum R_i Y_i$ is the sum of ranks corresponding to positive values of $X_i - \theta_0$.
- Let $r_1 < \dots < r_y$ be the ranks corresponding to positive differences; vector of signs $(r_1, \dots, r_n)$.
- $Y \sim \text{Bin}(n, 0.5)$.
- Under H_0 , the vector of signs $(Y_1, \dots, Y_n)$ can take any of the 2^n values (from $00 \dots 0$ to $11 \dots 1$) with equal probability $\frac{1}{2^n}$.

1.4.4 Exact Null Distribution

Each configuration of signs gives a value for T . We count how many configurations sum to K to find $P(T = K)$.

Example: Exact Distribution, $n = 7$: Finding $P(T = 9)$

Find all subsets of ranks $\{1, 2, 3, 4, 5, 6, 7\}$ that sum to 9:

Subsets of length 2: $(2, 7), (3, 6), (4, 5)$

Subsets of length 3: $(1, 2, 6), (1, 3, 5), (2, 3, 4)$

Total combinations = 6.

$$P(T = 9) = \frac{6}{2^7}$$

Example: Other Probabilities for $n = 7$

$$P(T = 0) = \frac{1}{2^n} \quad (\text{empty set, all signs 0})$$

$$P(T = 1) = \frac{1}{2^n} \quad (\text{set } \{1\})$$

$$P(T = 2) = \frac{1}{2^n} \quad (\text{set } \{2\})$$

$$P(T = 3) = \frac{2}{2^n} \quad (\text{sets } \{3\} \text{ and } \{1, 2\})$$

1.4.5 Comparison of Tests

Test	Assumption	Use
t -test	$F \equiv \mathcal{N}$	One-sample centre
Sign test	F is continuous	One-sample centre
Signed Rank test	F is symmetric & continuous	One-sample centre

All three tests maintain the **same significance level** α .

Note

- If the assumption of **symmetry holds**, the Signed Rank test is more powerful than the Sign test.
- If the assumption of **symmetry does not hold**, the p-value obtained from the Signed Rank test will be *incorrect*.

1.5 Large-Sample Approximation for the Signed Rank Test

Theorem: Distribution Parameters for T

$$\mathbb{E}[T] = \frac{n(n+1)}{4}$$

$$\text{Var}(T) = \frac{n(n+1)(2n+1)}{24}$$

Standardised Statistic:

$$T^* = \frac{T - \mathbb{E}[T]}{\sqrt{\text{Var}(T)}} \approx \mathcal{N}(0, 1)$$

1.5.1 Derivation of Mean and Variance

Let $U_i = R_i Y_i$. The ranks are fixed numbers $1, \dots, n$ but Y_i is random. We view this as:

$$T = \sum_{i=1}^n U_i$$

Define V_i as the contribution of the i -th rank to the total sum T :

$$V_i = \begin{cases} i & \text{with probability } \frac{1}{2} \quad (\text{sign is positive}) \\ 0 & \text{with probability } \frac{1}{2} \quad (\text{sign is negative}) \end{cases}$$

Abstract interpretation: We define V_i as the contribution of the i -th rank to the total sum T . Instead of summing randomised ranks, we look at the fixed rank value i (e.g., rank 1, rank 2, rank 5) and ask: *Will this rank be added to our total?*

- If the sign is positive: we add the rank value i to our total (Value = i).
- If the sign is negative: we do not add anything (Value = 0).

Since the sign is a “coin flip” (50/50 chance) under H_0 : V_i takes the value i with probability $1/2$; V_i takes the value 0 with probability $1/2$.

$$\mathbb{E}[V_i] = i \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{i}{2}$$

$$\mathbb{E}[T] = \sum_{i=1}^n \frac{i}{2} = \frac{1}{2} \cdot \frac{n(n+1)}{2} = \frac{n(n+1)}{4}$$

$$\text{Var}(V_i) = \mathbb{E}[V_i^2] - (\mathbb{E}[V_i])^2 = \frac{i^2}{2} - \frac{i^2}{4} = \frac{i^2}{4}$$

$$\text{Var}(T) = \sum_{i=1}^n \frac{i^2}{4} = \frac{1}{4} \sum_{i=1}^n i^2 = \frac{1}{4} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(2n+1)}{24}$$

Note

- V_i are independent because Y_i are independent.
- This approximation is useful when n is moderately large.
- Computation of 2^n possibilities is very time-consuming for large n .

1.6 Handling Ties

1.6.1 Case 1: Zeros

- If $X_i - \theta_0 = 0$ for some i .
- **Method:** Drop those observations and reduce n .

1.6.2 Case 2: Equal Absolute Differences

- If $|X_i - \theta_0| = |X_j - \theta_0|$ for $i \neq j$.
- **Method:** Assign the **average rank**.

Example: Average Rank Calculation

Values to rank: 5, 6, 5, 6, 7.

1. Sort them: 5, 5, 6, 6, 7.

2. Initial positions: 1, 2, 3, 4, 5.

3. Average ties:

- The two 5s are at positions 1 and 2. Average rank = $\frac{1+2}{2} = 1.5$.
- The two 6s are at positions 3 and 4. Average rank = $\frac{3+4}{2} = 3.5$.
- The 7 is at position 5. Rank = 5.

4. Final Ranks: 1.5, 1.5, 3.5, 3.5, 5.

Note

When ties are present and average ranks are used, the p-value derived is **approximate**, not exact.

1.7 R Code: Week 1

```
1  # This generates all possible subsets of size 2 when n = 4.
2  combn(c(1:4), 2)
3
4  # Writing own function for p-value of signed rank test
5  signrank <- function(n, T){
6    x <- c(1:n)
7    k <- 0
8    for(i in 1:n){
9      y <- combn(x, i)
10     l <- dim(y)[2]
11     for(j in 1:l){
12       k <- cbind(k, sum(x[y[,j]]))
13     }
14   }
15   pval <- length(which(k >= T)) / length(k)
16   return(pval)
17 }
18
19 table(k)
20
21 # Signed rank test for the example done in class
22 a <- c(73, 82, 87, 68, 106, 60, 97)
23 b <- a - 70
24 r <- rank(abs(b))
25 s <- ifelse(b > 0, 1, 0)
26 T <- sum(r * s)
27 signrank(7, T)
28
29 # Directly using built-in function for signed rank test
30 library(stats)
31 wilcox.test(a, mu = 70, alternative = "greater")
```

Listing 1.1: Generating subsets and computing the signed rank test p-value

Chapter 2

One-Sample Estimation and Two-Sample Tests

2.1 One-Sample Estimation of Centre

Last week we learned about tests for the centre of one sample the non-parametric equivalent of the t -test. Now we move to **estimation**. The point estimate of the population median can be taken as the sample median. The question is about *interval* estimation. We also know from general inference that hypothesis tests can be inverted to obtain confidence intervals.

2.1.1 Derivation of Interval Estimates

Model Setup

Let θ be the population median.

$$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F, \quad \text{med}(F) = \theta.$$

Hypothesis and Test Statistic

Given the null hypothesis $H_0 : \theta = a$, the test statistic Y is defined as:

$$Y = \#\{X_i < a\} \sim \text{Bin}(n, \frac{1}{2}) \quad \text{under } H_0.$$

Do not reject H_0 if:

$$\text{qbinom}(\frac{\alpha}{2}, n, 0.5) < Y < \text{qbinom}(1 - \frac{\alpha}{2}, n, 0.5).$$

Acceptance Region

The bounds for the acceptance region are determined by the quantiles of the Binomial distribution. The observed acceptance region is specified as:

$$179 < Y < 218.$$

Confidence Interval Construction

Using the count from the data: $\#\{X_i < a\} = 179$.

We invert the test to find the bounds for the parameter a :

$$a = X_{(179)}.$$

Thus the Confidence Interval (CI) is:

$$CI = (X_{(179)}, X_{(218)}).$$

Theorem: CI from Inverting the Test

$(X_{(L)}, X_{(U)})$ is the CI for a , where $(L < Y < U)$ is the acceptance region of the test.

2.1.2 Example: Academic Salary Data

Example: Faculty Salary Confidence Interval

The 200809 nine-month academic salary for Assistant Professors, Associate Professors, and Professors in a U.S. college. Data collected as part of the ongoing effort of the college's administration to monitor salary differences between male and female faculty members.

```
1 library(carData)
2 data("Salaries")
3 qbinom(0.025, 397, 0.5)           # gives 179
4 qbinom(0.975, 397, 0.5)           # gives 218
5 sort(Salaries$salary)[c(179, 218)] # gives (105000,
   111350)
```

What is the a so that k many X_i are less than a ? It is the k -th order statistic. So the confidence interval is $(X_{(179)}, X_{(218)})$.

When n is large, the `qbinom` quantities can be approximated by $n/2 \pm \Phi^{-1}(\alpha/2)\sqrt{n/4}$, since under H_0 , $\mathbb{E}(Y) = n/2$ and $\text{Var}(Y) = n/4$.

2.2 Two-Sample Test for Centre

Now we start testing for equality of centre of two samples the non-parametric equivalent of the two-sample t -test when the parent distributions are not normal and sample sizes are small.

2.2.1 Setup

Let $X_1, \dots, X_{n_1} \stackrel{\text{i.i.d.}}{\sim} D_1$ and $Y_1, \dots, Y_{n_2} \stackrel{\text{i.i.d.}}{\sim} D_2$.

Hypothesis:

$$H_0 : D_1 \equiv D_2 \quad (\text{Distributions are identical})$$

Alternatives:

$$H_a : D_1 \text{ is shifted to the right of } D_2 \quad (\text{right-tailed})$$

$$H_a : D_1 \text{ is shifted to the left of } D_2 \quad (\text{left-tailed})$$

$$H_a : D_1 \text{ is shifted right or left of } D_2 \quad (\text{two-sided})$$

Assumptions:

- Continuous distributions.
- Same shape (shift model).
- Independence of data values.
- Samples must be random and independent.

2.3 Wilcoxon Rank Sum Test

Definition: Wilcoxon Rank Sum Test

A non-parametric test for comparing the centres of two independent samples, using ranks of the pooled sample. Without loss of generality, let $n_2 < n_1$.

2.3.1 Procedure

Procedure: Wilcoxon Rank Sum Algorithm

1. Pool together X 's and Y 's.
2. Order the pooled sample and assign ranks.
3. Find the sum X of ranks of the X_i 's in the pooled sample.
4. The test statistic is $T = X - \frac{n_1(n_1 + 1)}{2}$.

Logic:

- If X 's are larger than Y 's (right-tailed alternative), then D_1 is shifted to the right of D_2 .
- Consequently, the *average ranks* of X 's will be larger than the average ranks of Y 's.

Test Statistic:

$$T = \text{sum of ranks of } X_i\text{'s in the pooled sample} - \frac{n_1(n_1 + 1)}{2}$$

$$T + T_1 = \frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2} - \frac{n_1(n_1 + 1)}{2} = C$$

where T_1 is the sum of ranks of Y_i 's. Therefore:

$$T_1 = C - T.$$

2.3.2 Numerical Example: Reaction Times

Example: Reaction Time Under Drugs A and B

We compare reaction times under the influence of two drugs (or treatments).

$$X = (2.07, 2.11, 2.43, 2.50, 2.71, 2.84, 2.88), \quad n_1 = 7$$

$$Y = (1.62, 1.71, 1.93, 1.96, 2.24, 2.41), \quad n_2 = 6$$

Ranks after pooling and sorting:

- Y's occupy ranks: 1, 2, 3, 4, 7, 8.
- X's occupy ranks: 5, 6, 9, 10, 11, 12, 13.

Statistic Calculation:

$$T = 5 + 6 + 9 + 10 + 11 + 12 + 13 = 66$$

$$U = 66 - \frac{n_1(n_1 + 1)}{2} = 66 - \frac{7 \times 8}{2} = 66 - 28 = 38.$$

Result: P-value = 0.01399 (two-tailed). At $\alpha = 5\%$, we **reject** H_0 , i.e., $D_1 \neq D_2$. With $T = 38$, $m = 7$, $n = 13$ in R:

```
1 # The ranks of Y in the pooled sample are 5, 6, 9, 10,
   # 11, 12, 13.
2 # So T = 38, m = 7, n = 13.
3 # The p-value obtained from R is 0.01399.
```

2.3.3 Null Distribution Construction

How do we find the distribution of the test statistic under H_0 ($D_1 \equiv D_2$)?

Procedure: Permutation Method for Null Distribution

1. Generate all combinations of size n_1 from the pooled set of size $n_1 + n_2$.
2. Total number of combinations: $\binom{n_1 + n_2}{n_1}$.
3. For each combination, label as "Group X" and calculate T .
4. Since H_0 implies exchangeability, all combinations are equally likely (probability = $1/\binom{n_1 + n_2}{n_1}$).
5. This generates the "Null Distribution".
6. Find the position of the observed T in this distribution to get the p-value.

2.4 Permutation Tests

Definition: Permutation Tests

Tests that use permutations to obtain the null distribution. This technique can be applied to *any* test statistic even parametric statistics without requiring distributional assumptions.

2.4.1 Application to the t -Statistic

Consider the two-sample t -statistic:

$$t = \frac{\bar{x} - \bar{y}}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

We do not know the distribution of this statistic unless X 's and Y 's are Normal. However, we can generate the distribution using the Permutation Technique.

Procedure: Permutation t -Test Procedure

1. Shuffle the data labels.
2. Re-calculate the t -statistic for every shuffle.
3. This empirical distribution serves as the null distribution, allowing the test without assuming Normality.

Remark

There is no reason whatsoever why a permutation test has to use any particular test statistic. **Any test statistic will do!**

2.5 R Code: Week 2

```

1  # Wilcoxon rank sum test
2  # Example: Hollander & Wolfe (1973)
3  # Permeability constants of the human chorioamnion (a
4  # placental membrane) at
5  # term (x) and between 12 to 26 weeks gestational age (y).
6  # Alternative: greater permeability at term.
7  x <- c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91, 1.64, 0.73,
8  #      1.46)
9  y <- c(1.15, 0.88, 0.90, 0.74, 1.21)
10 z <- sort(c(x, y))
11 T <- sum(which(z %in% x)) - 10*11/2
12
13 ranksum <- function(n, m, T){
14   T <- T + m*(m+1)/2
15   x <- c(1:n)

```

```
15 l <- choose(n, m)
16 k <- rep(0, l)
17 y <- combn(x, m)
18 for(j in 1:l){
19   k[j] <- sum(x[y[,j]])
20 }
21 pval <- 2 * length(which(k < T)) / choose(n, m)
22 pval <- ifelse(pval > 1, 2 - pval, pval)
23 return(pval)
24 }
25
26 ranksum(15, 10, 35)
27
28 wilcox.test(
29   c(0.80, 0.83, 1.89, 1.04, 1.45, 1.38, 1.91, 1.64, 0.73,
30     1.46),
31   c(1.15, 0.88, 0.90, 0.74, 1.21)
32 )
33 # Permutation test using t-statistic
34 T <- t.test(x, y)$statistic
35
36 t.pval <- function(a, b, T){
37   c <- cbind(a, b)
38   m <- length(b)
39   n <- length(c)
40   x <- c(1:n)
41   l <- choose(n, m)
42   t <- rep(0, l)
43   y <- combn(x, m)
44   for(j in 1:l){
45     t[j] <- t.test(c[y[,j]], c[-y[,j]])$statistic
46   }
47   pval <- 2 * length(which(t >= T)) / l
48   pval <- ifelse(pval > 1, 2 - pval, pval)
49   return(pval)
50 }
51
52 t.pval(x, y, T)
```

Listing 2.1: Wilcoxon Rank Sum Test and Permutation Test

Chapter 3

Mann-Whitney U-Statistic, General Score Functions, and Scale Tests

3.1 Wilcoxon Rank Sum and Mann-Whitney Equivalence

Definition: Wilcoxon Rank Sum Statistic

$$T = \sum_{i=1}^{n_1} R_{X_i} - \frac{n_1(n_1 + 1)}{2}$$

where R_{X_i} is the rank of X_i in the pooled sample $\{X_1, \dots, X_{n_1}, Y_1, \dots, Y_{n_2}\}$.

Definition: Mann-Whitney U-Statistic

$$T^* = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbf{1}(X_i > Y_j)$$

Theorem: Equivalence of T and T^*

$$T = T^*$$

Derivation:

$$T = \sum_{i=1}^{n_1} (R_{X_i} - i) = \sum_{i=1}^{n_1} \#\{y_j < x_{(i)}\}$$

Rank Definition Logic:

$$R_{X_{(i)}} = \#\{\text{samples} < x_{(i)}\} = i + \#\{y_j < x_{(i)}\}$$

Thus $T = T^*$, where T^* is the Mann-Whitney U-statistic.

3.1.1 Properties of the U-Statistic

- The U-statistic is a **sum of unbiased estimators**.
- Let $\theta = \mathbb{E}[\mathbf{1}(X_{(i)} > Y_{(j)})] = P(X > Y)$ for some parameter θ .

- $\mathbb{E}[T^*] = n_1 n_2 \theta$.

Under the Null Hypothesis (H_0):

$$\theta = \frac{1}{2}$$

$$\mathbb{E}[\hat{\theta}] = \frac{n_1 n_2}{2}$$

$$\text{Var}(\hat{\theta}) = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

3.1.2 Asymptotic Normality

A general procedure exists for finding the variance of U-statistics and establishing asymptotic normality:

$$Z = \frac{T^* - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}} \xrightarrow{d} \mathcal{N}(0, 1)$$

(This is the two-sample test for centre using the large-sample approximation.)

3.2 General Linear Rank Statistics

3.2.1 General Form

Definition: General Linear Rank Statistic

The general form of rank statistics is:

$$\alpha \sum R_i + \beta.$$

More specifically, the **General Linear Rank Statistic** T_1 is:

$$T_1 = \sum_{i=1}^{n_1} \varphi\left(\frac{R_i}{N+1}\right)$$

where $N = n_1 + n_2$ is the total sample size and $\varphi(u)$ is a **score function** defined on $(0, 1)$.

3.2.2 Assumptions on the Score Function φ

- $\int_0^1 \varphi(u) du = 0 \implies \mathbb{E}[T_1] = 0$
- $\int_0^1 \varphi^2(u) du = 1 \implies \text{Var}(T_1) \approx 1$ (scale normalisation)

For general scores, we define the process:

$$S_\varphi(\Delta) = \frac{1}{n} \sum_{j=1}^{n_2} a_\varphi[R(Y_j - \Delta)]$$

where $R(Y_j - \Delta)$ denotes the rank of $Y_j - \Delta$ among $X_1, \dots, X_{n_1}$ and $Y_1 - \Delta, \dots, Y_{n_2} - \Delta$. A test statistic for $H_0 : \Delta = 0$ vs $H_A : \Delta > 0$ is $S_\varphi(0)$.

The generated scores are $a_\varphi(i) = \varphi[i/(n+1)]$, where $n = n_1 + n_2$. The Wilcoxon score function is $\varphi_W(u) = \sqrt{12}[u - \frac{1}{2}]$.

The mean and variance of $S_\varphi(0)$ under the null hypothesis are:

$$\bar{a} = \frac{1}{n} \sum_{i=1}^n a_\varphi(i), \quad \frac{n_1 n_2}{n(n-1)} \sum_{i=1}^n (a_\varphi(i) - \bar{a})^2.$$

3.3 Measure of Scale

Remark

There is no unique measure of scale for a general distribution (unlike the median, which is the non-parametric alternative to the mean).

A general measure of scale for a distribution is:

$$(\mathbb{E}[|X - \mu|^k])^{1/k}, \quad M = \text{Median}.$$

3.4 Test for Comparing Scales of Two Populations

3.4.1 Setup

- $X_1, \dots, X_{n_1} \sim F_X$
- $Y_1, \dots, Y_{n_2} \sim F_Y$

Location-Scale Model:

$$\text{pdf of } X = f_X\left(\frac{x - \mu_1}{\sigma_1}\right), \quad \text{pdf of } Y = f_Y\left(\frac{x - \mu_2}{\sigma_2}\right).$$

Same form f , different centre and scale. Let $n_1 = 21$ (control) and $n_2 = 22$ (experimental).

Hypothesis:

$$H_0 : \sigma_1 = \sigma_2 \quad H_a : \sigma_1 > \sigma_2.$$

3.4.2 Fligner-Killeen (FK) Test

Procedure: Fligner-Killeen Test Strategy

1. Estimate μ_1, μ_2 using sample medians.
2. Calculate absolute deviations (aligned samples):

$$u_i = |x_i - \hat{\mu}_1|, \quad v_j = |y_j - \hat{\mu}_2|.$$

That is: $X_i^* = X_i - \text{med}_X$ and $Y_j^* = Y_j - \text{med}_Y$.

3. If $\sum u_i \geq \sum v_j$, then u_i is shifted relative to v_j .
4. Perform **Wilcoxon Rank Sum Test** on $\{u_i\}$ and $\{v_j\}$.

Remark

Any monotonic increasing function can be used for the test. General rank statistics can be applied here with a different φ from the linear one. Equivalent to taking squares, absolute values, or $\log |\cdot|$.

A more common test is the **Fligner-Killeen (FK)** procedure using the normal score function:

$$\varphi_{FK}(u) = \left(\Phi^{-1} \left(\frac{u + 1}{2} \right) \right)^2$$

where Φ is the standard normal CDF. This leads to the test statistic:

$$S_{FK} = \sum_{j=1}^{n_2} \left(\Phi^{-1} \left(\frac{R|Y_j^*|}{2(n+1)} + \frac{1}{2} \right) \right)^2.$$

3.4.3 Numerical Result

Example: Fligner-Killeen Test: Ozone Effect on Rats

Draper and Smith (1966) present an experiment on the effect of ozone on the weight gain of rats. Two groups of rats were selected.

- **Control group** ($n_1 = 21$): ozone-free environment for 7 days.
- **Experimental group** ($n_2 = 22$): ozone environment for 7 days.

At the end of 7 days, the gain in weight of each rat was recorded.

$$R_i = \text{rank of } u_i = |x_i - \hat{\mu}_1| \text{ in the pooled sample}, \quad S_\varphi = \sum_{i=1}^{n_1} \varphi(R_i).$$

Results:

- Statistic $Z = 2.095$
- P-value = 0.036 (one-tailed)
- **Conclusion:** Reject H_0 (one-sided test required).

Confidence Interval for the ratio of scales σ_2/σ_1 :

$$\text{CI} = (1.002, 5.63).$$


```
1 > data(sievers)
2 > x <- with(sievers, weight.gain[group == 'Control'])
3 > y <- with(sievers, weight.gain[group == 'Ozone'])
4 > fk.test(x, y)
5 statistic = 2.095976 , p-value = 0.03608434
6 95 percent confidence interval:
7 1.002458 5.636436
8 Estimate: 2.377034
```


Chapter 4

Multiple Samples, CDF Inference, and Goodness of Fit

4.1 Multiple Samples

4.1.1 Parametric Case

We have m samples drawn from m populations:

$$\begin{aligned} X_{11}, \dots, X_{1n_1} &\stackrel{\text{i.i.d.}}{\sim} F_1(\mu_1) \\ X_{21}, \dots, X_{2n_2} &\stackrel{\text{i.i.d.}}{\sim} F_2(\mu_2) \\ &\vdots \\ X_{m1}, \dots, X_{mn_m} &\stackrel{\text{i.i.d.}}{\sim} F_m(\mu_m) \end{aligned}$$

(m samples from m populations)

$$H_0 : \mu_1 = \dots = \mu_m.$$

Note

ANOVA requires the assumption that F_i are normal.

4.1.2 Non-Parametric Case

For general F_i testing for equality of the sample **medians**.

Procedure: Kruskal-Wallis Procedure

1. Put all X_{ij} 's together and **rank** the combined sample.
2. For sample j ($j = 1, \dots, m$), take the corresponding ranks (elements coming from the j -th sample).
3. Sum those ranks: R_j , $j = 1, \dots, m$.

Under H_0 :

$$\frac{R_j}{n_j} \longrightarrow \text{these should be close.}$$

There should be **no difference** (or very little) between the average ranks of different groups.

The **test statistic** is the variance of R_j/n_j :

$$\sum_j n_j \left(\frac{R_j}{n_j} - \bar{R} \right)^2$$

where $n = \sum_{j=1}^m n_j$ and

$$\bar{R} = \text{avg rank for all samples together} = \frac{n(n+1)}{2n} = \frac{n+1}{2}.$$

4.2 Kruskal-Wallis Test

Theorem: Kruskal-Wallis Statistic

$$\begin{aligned} H &= \frac{12}{n(n+1)} \left[\sum_{j=1}^m \frac{R_j^2}{n_j} - \frac{(n+1)^2 n}{4} \right] \\ &= \frac{12}{n(n+1)} \sum_{j=1}^m \frac{R_j^2}{n_j} - 3(n+1). \end{aligned} \quad (4.1)$$

$H \approx \chi_{m-1}^2$ (approximately); n_j should be large for each j .

The exact distribution of the test statistic can be derived by **permutation**.

$$n = n_1 + n_2 + \cdots + n_m, \quad H_1, \dots, H_K.$$

Number of permutations:

$$\frac{n!}{n_1! n_2! \cdots n_m!} \quad (\text{many permutations})$$

Example:

$$\binom{14}{5} \binom{9}{7} = \frac{14!}{5! 7! 2!}$$

1. Calculate H for each permutation.
2. Find the p-value from the position of the observed H in the histogram.

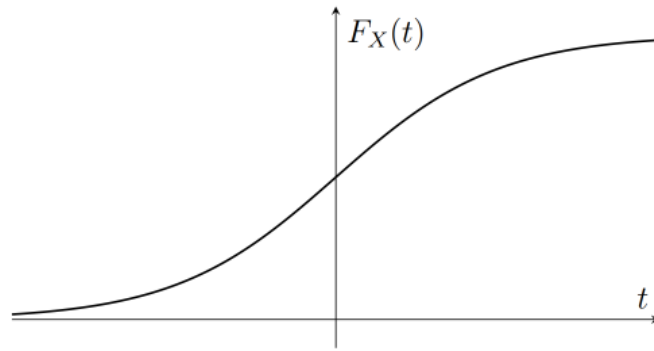


Figure 4.1

Example: Kruskal-Wallis: Mucociliary Efficiency**Reference:** Hollander & Wolfe (1999).**Context:** Foam rate of diet in $m = 3$ groups:

- Normal
- Obstructive airway
- Asbestosis

 $n_j : 5, 4, 5; \quad n = 14.$ **HW:** Write R code for this problem.

4.3 Inference for the CDF

Definition: CDF as an Infinite-Dimensional Parameter

The family $\{F_x(b), -\infty < b < \infty\}$ is the parameter an **infinite-dimensional** parameter.

$X \rightarrow$ random variable; $F_x(t) = P(X \leq t)$; $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F_x$.

For $t = 5$: $P(X \leq 5) = \theta$ is estimated by $\frac{1}{n} \#\{x_i \leq 5\}$, giving estimate $\hat{\theta}$.

4.3.1 Empirical CDF

Definition: Empirical CDF

$$\hat{F}_n(t) = \frac{1}{n} \#\{x_i \leq t\}$$

For $\theta = F_x(t)$, a point estimate is $\hat{\theta} = \hat{F}_n(t)$.

Distribution of $\hat{\theta}$:

$$Y(t) = \#\{x_i \leq t\} = \sum_{i=1}^n \mathbf{1}(x_i \leq t) \sim \text{Bin}(n, \theta), \quad P(x_i \leq t) = F_x(t) = \theta.$$

If n is large enough, we can use the CLT for the Binomial.

CI for θ :

$$\left(\hat{\theta} \pm z_{\alpha/2} \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} \right)$$

$$\hat{\theta} = \frac{Y(t)}{n} = \frac{1}{n} \# \{x_i \leq t\}.$$

Note

The estimate of F is a **step function** (discontinuous), even when F is continuous.

4.3.2 Order Statistics and the Empirical CDF

Given the order statistics $X_{(1)} < \dots < X_{(n)}$:

$$\hat{F}_n(t) = \begin{cases} 0 & t < X_{(1)} \\ \frac{1}{n} & X_{(1)} \leq t < X_{(2)} \\ \frac{2}{n} & X_{(2)} \leq t < X_{(3)} \\ \vdots & \\ \frac{n-1}{n} & X_{(n-1)} \leq t < X_{(n)} \\ 1 & X_{(n)} \leq t \end{cases}$$

Changes at $n + 1$ points. No need to compute for every t .

CDF, PDF; quantile function $F^{-1}(x)$. Quantile β^{th} : $\{F^{-1}(x)\}$, $0 \leq x \leq 1$ a parameter.

4.3.3 Testing / Estimation of Median

When $x = \frac{1}{2}$: $\theta = \text{median}$; $\hat{\theta} = \text{sample median}$.

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_a : \theta \neq \theta_0.$$

$$\frac{\hat{\theta} - \theta_0}{\sqrt{\frac{\theta_0(1-\theta_0)}{n}}} \sim \mathcal{N}(0, 1).$$

Also derive an **exact test** for the median.

When $x = \frac{3}{4}$, the third percentile = $\hat{\theta}$: $\hat{\theta} = \text{sample 3rd quartile}$.

$$\hat{\theta} = \min\{x : 0.75n \text{ of the } x_i\text{'s are } \leq x\}.$$

CI and test for quantiles.

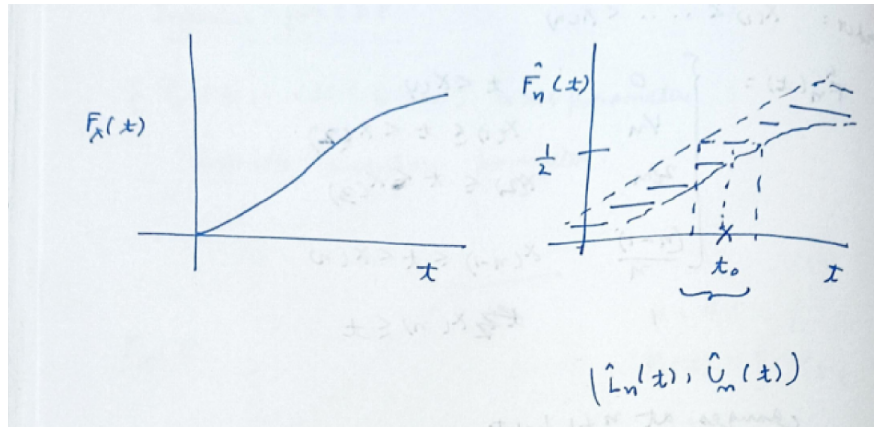


Figure 4.2: CDF and its Confidence interval

4.4 Confidence Intervals for CDF and Quantile Functions

4.5 Testing for Median (L-6)

4.5.1 Is the Population Median 0.2?

Sample $X_1, \dots, X_n$.

$$H_0 : P(X > 0.2) = \frac{1}{2} = p_0 \quad H_a : p \neq p_0.$$

$$\hat{p} = \frac{1}{n} \# \{x_i > 0.2\}.$$

Under H_0 : $n\hat{p} \sim \text{Bin}(n, \frac{1}{2})$.

Either **binomial test** or **normal approximation** if n is sufficiently large.

4.5.2 Is the Third Quartile 0.3?

$$H_0 : P(X > 0.3) = 0.25 = p_0 \quad \text{or equivalently} \quad H_0 : P(X < 0.3) = 0.75.$$

$$H_a : p \neq p_0, \quad p = P(X > 0.3).$$

$$\hat{p} = \frac{1}{n} \# \{x_i > 0.3\}.$$

Under H_0 : $n\hat{p} \sim \text{Bin}(n, 0.25)$.

P-value or critical region; use exact binomial or normal approximation.

Example: Testing the Third Quartile

Given 176 observations out of 250 are less than $a = 1.5$. Test if 1.5 is the third quartile.

$$p = P(X < 1.5), \quad H_0 : p = 0.75, \quad H_a : p \neq 0.75, \quad \alpha = 5\%.$$

$$\hat{p} = \frac{176}{250}.$$

$$n\hat{p} \sim \text{Bin}(n, p_0) = \text{Bin}(250, 0.75).$$

$$Z = \frac{176 - 250 \times 0.75}{\sqrt{250 \times 0.75 \times 0.25}} = -1.68.$$

$$\text{p-value from } (-1.68) = 0.0929.$$

We do **not** reject at the 5% α -level.

$$2 \times \text{pbinom}(176, 250, 0.75) = 0.1119 \quad (\text{exact}).$$

4.6 Estimating the CDF Further Details

Estimate CDF: $F_x(t) = P(X \leq t) = \theta$.

$$\hat{\theta} = \frac{1}{n} \# \{x_i \leq t\}.$$

$$n\hat{\theta} \sim \text{Bin}(n, \theta).$$

CI for θ :

$$\left(\hat{\theta} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} \right).$$

$$\hat{F}_n(t) = \begin{cases} 0 & t < X_{(1)} \\ 1/n & X_{(1)} \leq t < X_{(2)} \\ \vdots & \\ k/n & X_{(k)} \leq t < X_{(k+1)} \\ 1 & t > X_{(n)} \end{cases}$$

Not continuous; each step height is $\frac{1}{n}$.

4.7 Goodness of Fit: χ^2 and KS Tests

Testing goodness of fit of a distribution.

$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F; H_0 : F = F_0.$

4.7.1 Case 1: F_0 Discrete χ^2 Test

$$\chi^2 = \sum \frac{(\text{Obs} - \text{Exp})^2}{\text{Exp}}$$

Binning: histogram / frequency table. For continuous distributions, there is no obvious / objective choice of binning. Each bin has the same probability under H_0 .

4.7.2 Kolmogorov-Smirnov Statistic

Definition: Kolmogorov-Smirnov (KS) Statistic

$$D = \max_{-\infty < t < \infty} |\hat{F}_n(t) - F_0(t)| = \max_{t=X_{(1)}, \dots, X_{(n)}} \{|\hat{F}_n(t) - F_0(t)|\}.$$

The maximum occurs at one of the data points; max step size = $\frac{1}{n}$.

Theorem: Properties of the KS Statistic

1. Evaluation is taken at **finitely many** points.
2. D is **distribution-free**.

4.7.3 Distribution-Free Property of D

$X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F_0$. Continuous distribution F_0 increasing: $F(X) \sim \text{Unif}(0, 1)$.

Let $S = \hat{F}_n(t) - F_0(t)$:

$$\begin{aligned} S &= \frac{1}{n} \# \{x_i \leq t\} - F_0(t) \\ &= \#(i : F_0(x_i) \leq F_0(t)) \cdot \frac{1}{n} - F_0(t) \\ &= \#(i : U_i \leq F_0(t)) \cdot \frac{1}{n} - F_0(t), \quad U_1, \dots, U_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1) \\ &= \#(i : U_i \leq u) \cdot \frac{1}{n} - u, \quad [F_0(t) = u] \\ &= \tilde{F}_n(u) - u \end{aligned}$$

where $\tilde{F}_n$ is the ECDF of $U_1, \dots, U_n$.

Theorem: Distribution-Free Result

The RHS does *not* depend on F_0 , so D is **distribution-free**. It suffices to derive the distribution of D when $F_0 = \text{Unif}(0, 1)$.

Chapter 5

Density Estimation

5.1 Density Estimation

$f_x \rightarrow$ parameter (infinite dimensional parameter) of interest

1. $f_x \geq 0$
2. $\int f(x) dx = 1$
3. $\int_a^b f(x) dx = P(a < X < b)$

$X_1, \dots, X_n \sim f_x$ have distribution same as X .

$\hat{f} \rightarrow$ histogram. Area under the curve as integral, then indeed $\int \hat{f}(x) dx = 1$.

$\hat{f} \geq 0$

For a, b endpoints of a bin,

$$\int_a^b \hat{f}(x) dx = \frac{\#\{X_i \in (a, b)\}}{n} = \hat{F}_n(b) - \hat{F}_n(a) = \text{empirical probability.}$$

Histogram satisfies (1)–(3).

Problems

1. It is not smooth.
2. Location and width of bins is subjective. (not a function of data only).

Minimize MSE to find optimal bin width (bandwidth).

Find MSE. $x_0, x_1, x_2, \dots, x_n \rightarrow$ endpoints of bins.

$$\begin{aligned} \hat{f}(t) &= \begin{cases} \frac{\hat{F}_n(x + \delta) - \hat{F}_n(x)}{\delta} & x_i < t < x_{i+1} \\ 0 & \text{otherwise} \end{cases} \\ x_i &= x, \quad x_{i+1} = x + \delta \\ \hat{f}(t) &= \frac{\hat{F}_n(x + \delta) - \hat{F}_n(x)}{\delta} = \frac{\#\{X_i \in (x, x + \delta)\}}{n\delta} \end{aligned}$$

$$n\hat{F}_n(x) \sim \text{Bin}(n, F(x)) \quad \sim \frac{1}{n\delta} \text{Bin}(n, F(x + \delta) - F(x))$$



Figure 5.1: Step Function Diagram

$$F(x + \delta) - F(x) = p$$

$$E\hat{f}(t) = \frac{1}{n\delta} \cdot n[F(x + \delta) - F(x)] = \frac{1}{\delta}[F(x + \delta) - F(x)] = \frac{p}{\delta}$$

$$\text{Var } \hat{f}(t) = \frac{1}{n^2\delta^2} np(1-p) = \frac{p(1-p)}{n\delta^2}$$

$$\text{MSE} = \text{Bias}^2 + \text{Var} = \left(\frac{p}{\delta} - f(t)\right)^2 + \frac{p(1-p)}{n\delta^2}$$

Minimize thus over δ to get bin width.

$$\left[\begin{array}{l} \text{Best choice is } \delta \propto n^{-1/5}. \text{ Larger sample size should have} \\ \text{denser (more bins) histograms.} \\ \text{MSE} = O(n^{-4/5}) \text{ not } 1/n. \quad \text{MSE}(x) = \frac{\sigma^2}{n} \end{array} \right.$$

Location – Shifted histogram. (fix δ).

Actual evaluation does not need to be done at every ($\mathbb{R}$) points, but only at discrete set of points.

$$\hat{f}(t) = \#\{X_i \in (t - \delta/2, t + \delta/2)\} / \delta$$

Smoothness: Kernel density.

$K(z)$ is a standard kernel function if

- K is Symmetric around zero $k(z) = K(-z)$
- $K(z) \geq 0 \forall z \quad \int_{-\infty}^{\infty} K(z) dz = 1$
- The mass function is non decreasing on both sides of zero.
- $\int z k(z) dz = 0 \quad \int z^2 k(z) dz = 1$

Bandwidth re-scales kernel

$$k_h(z) = \frac{1}{h} K\left(\frac{z}{h}\right)$$

e.g. Rectangular $K(z) = \frac{1}{2}\mathbf{1}(-1 < z < 1)$



Figure 5.2: smooth bump-shaped density estimate over data points $x_1, x_2, \dots$

Gaussian $K(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$

Triangular $K(z) = (1 - |z|) \mathbf{1}(-1 < z < 1)$

Epanechnikov $K(z) = \frac{3}{4}(1 - z^2) \mathbf{1}(-1 < z < 1)$

Kernel density estimators

$$\hat{f}_h(t) = \frac{1}{n} \sum_{i=1}^n k_h(X_i - t)$$

Shifted histogram is a kernel density estimate with rectangular kernel. The height of the shifted histogram at a point t is proportional to the # of data points at a distance less than h to t .

Convolution of two densities f_x, g_y

$$f_S(z) = f_x * g_y(z) = \int_{-\infty}^{\infty} f_x(t) g_y(z - t) dt = \int_{-\infty}^{\infty} f_x(z - t) g_y(t) dt$$

f_S is the density of a r.v. $S = X + Y$.

Convolution of 2 pmfs p_x and p_y

$$p_S(x) = p_x * p_y(x) = \sum_{t \in \mathbb{R}} p_x(t) p_y(x - t)$$

p_S is the pmf of $S = X + Y$

X independent of Y

f_x is density of X and p_y is the pmf of Y .

$S = X + Y$ is continuous.

$$P(S \leq a) = \sum_{k \in \mathbb{R}} P(Y = k \text{ and } X \leq a - k) = \sum_{k \in \mathbb{R}} p_y(k) F_x(a - k)$$

$$f_S(a) = \sum_{t \in \mathbb{R}} p_y(t) f_x(a - t)$$

Back to kernels.

$$\hat{f}(t) = \frac{1}{n} \sum_{i=1}^n k_h(t - X_i) = \sum_{i=1}^n p_y(X_i) k_h(t - X_i)$$

p_y is the pmf that puts mass $\frac{1}{n}$ at each $X_i, i = 1, \dots, n$.

$$p_y(t) = \begin{cases} \frac{1}{n} & \text{if } t \in \{x_1, \dots, x_n\} \\ 0 & \text{o.w.} \end{cases}$$

So if k_h is density of a cont r.v. then $\hat{f}(t)$ is also a density.

Bandwidth selection

Minimize MSE.

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n k_h(x - X_i) \quad k_h(t) = \frac{1}{h} k\left(\frac{t}{h}\right) \quad \int_{\mathbb{R}} k(u) du = 1$$

$$\begin{aligned} E\left(\hat{f}_h(x)\right) &= \frac{1}{n} \sum_{i=1}^n E(k_h(x - X_i)) \quad X_i \sim f \\ &= E(k_h(x - X_1)) \end{aligned}$$

$$\text{Var}(\hat{f}_h(x)) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(k_h(x - X_i)) = \frac{1}{n} \text{Var}(k_h(x - X_1))$$

$$E(k_h(x - X_1)) = \int k_h(x - t) f(t) dt \quad k_h(x) \simeq 0 \text{ for } |x| > h$$

$$\text{Taylor expansion} \quad f(t) = f(x) + (t - x) f'(x) + \frac{(t - x)^2}{2} f''(x) + o(h^2)$$

$$\begin{aligned} E(k_h(x - X_1)) &= \int k_h(x - t) f(x) dt + \int k_h(x - t) (t - x) f'(x) dt + \int k_h(x - t) \frac{(t - x)^2}{2} f''(x) dt \\ &= f(x) \cdot 1 + f'(x) \int (t - x) k_h(t - x) dt + \text{smaller terms} \\ &\quad + \frac{f''(x)}{2} \int (t - x)^2 k_h(t - x) dt + \text{smaller terms} \end{aligned}$$

$$E(\hat{f}|x) = f(x) + 0 + \frac{h^2}{2} f''(x) + o(h^2)$$

$$\text{Bias} = \frac{h^2}{2} f''(x) + o(h^2).$$

$$\begin{aligned} E(k_h^2(x - X_1)) &= \int k_h^2(x - t) f(t) dt \\ &= \int \left[\frac{1}{h} k\left(\frac{x - t}{h}\right) \right]^2 f(t) dt \\ &= \frac{1}{h^2} \int k^2\left(\frac{x - t}{h}\right) f(t) dt \quad \frac{x - t}{h} = u \\ &= \frac{1}{h^2} \cdot h \int k^2(u) f(x + hu) du \\ &= \frac{1}{h} \left[\int k^2(u) f(x) du + \text{smaller terms} \right] \end{aligned}$$

$$\begin{aligned} \text{Var}(\hat{f}_h(x)) &= \frac{1}{n} \left[\frac{1}{h} \cdot f(x) \cdot \int k^2(u) du + \text{smaller terms} - (E\hat{f})^2 \right] \\ &= \frac{1}{nh} \int f(x) \cdot k^2(u) du + \text{smaller terms.} \end{aligned}$$

$$\text{Bias} = \frac{h^2}{2} f''(x)$$

$$\text{MSE} = h^4 C_1(x) + \frac{1}{nh} C_2(x) + \text{smaller terms.}$$

$$h^4 = O\left(\frac{1}{nh}\right) \quad \text{To minimize MSE}$$

$$h \propto n^{-1/5}$$

$$h^4 = n^{-4/5}$$

$$\text{MSE} = O(n^{-4/5})$$

$$\frac{1}{nh} = n^{-1}h^{-1} = n^{-4/5}$$

$h < O(n^{-1/5})$ then $\frac{1}{nh}$ is of higher order than $n^{-4/5}$

$h > O(n^{-1/5})$ then h^4 is of higher order than $n^{-4/5}$

Normal reference rule.

If f is normal then

$$h = 1.06 \sigma n^{-1/5}$$

= One can estimate σ from the data.

If the histogram is symmetric unimodal then this choice of h can be used.

Cross-validation to determine h

$$E\left(\hat{f}_h(x) - f(x)\right)^2$$

minimise $\int \left(\hat{f}_h(x) - f(x)\right)^2 dx$.

$$\begin{aligned} &= \int \hat{f}_h^2(x) dx - 2 \int \hat{f}_h(x) f(x) dx + \int f^2(x) dx \\ &= F_1 + F_2 + F_3 \end{aligned}$$

For a particular h , we can compute $\hat{f}_h(x)$ for every x and hence approximate the first term.

The third term does not contain h , so it does not affect the choice of h .

Second term involves h (so we do need it) and the unknown f .

$$-2 \int \hat{f}_h(x) f(x) dx = -2 E \hat{f}_h(X) \quad \text{where } X \sim f.$$

You have $X_1, \dots, X_n$ i.i.d. f .

Estimate $E \hat{f}_h(x)$ by $\frac{1}{n} \sum_{i=1}^n \hat{f}_h(X_i)$.

Then minimise the approximation to F_1 (numerically) and (the estimate of F_2) (numerically) to get h .

$$\boxed{\frac{1}{n} \sum_{i=1}^n \hat{f}_h(X_i)} \rightarrow \text{is biased for } E\hat{f}_h(x)$$

$$\frac{1}{n} \sum_{i=1}^n \frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j)$$

$$E(k_h(X_i - X_j)) = E\hat{f}_h(X_i) \quad \text{for } i \neq j$$

$$i = j \quad E(k_h(X_i - X_j)) = E(k_h(0)) = k_h(0).$$

$$E \left[\frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j) \right] = \frac{1}{n} \left[(n-1) E\hat{f}_h(x) + k_h(0) \right] = E\hat{f}_h(X_i) + \frac{1}{n} \left(k_h(0) - E\hat{f}_h(x) \right)$$

$$E \left[\frac{1}{n} \sum_{j=1}^n k_h(X_i - X_j) \right] = \frac{1}{n} \left[(n-1) E\hat{f}_h(x_i) + k_h(0) \right] = E\hat{f}_h(X_i) + \underbrace{\frac{1}{n} \left(k_h(0) - E\hat{f}_h(x) \right)}_{\text{bias.}}$$

Correct this by removing $k_h(X_i - X_i)$

$$\hat{f}_{-i}(X_i) = \frac{1}{n-1} \sum_{j \neq i} k_h(X_i - X_j)$$

$$(-2) \frac{1}{n} \sum_{i=1}^n \hat{f}_{-i}(X_i) \rightarrow \text{estimates } F_2. \quad \text{Leave one out cross validation}$$